

■ Optimization & Decision Engine Engineer

Objective

Build the decision brain:

"Given the current situation and forecast, what should the farmer/FPO/buyer do?"
This is potentially the most innovative technical component.

What You Own

- Sell / Store / Split decision
- Risk-adjusted scoring
- Expected-return calculation
- Opportunity-cost calculation
- Storage-cost calculation
- Transportation-cost consideration
- Risk penalty
- Farmer risk tolerance modelling
- Recommendation ranking
- What-if simulation
- Decision rules
- Optimization algorithm

Questions You Should Answer / Investigate

- Linear programming?
- Integer programming?
- Constraint optimization?
- Expected utility?
- Risk-adjusted optimization?
- Multi-objective optimization?
- Heuristic approach?
- Note: Don't choose a complicated algorithm just to say "optimization." The objective should simply be the best decision under uncertainty and constraints.

Worked Example

Input:

Current price	= Rs. 2500
Predicted price	= Rs. 2800
Storage cost	= Rs. 120
Quantity	= 1000 kg
Risk tolerance	= Medium
Forecast confidence	= 78%

Engine determines:

SELL -> 30%
STORE -> 70%

What-if simulation:

Storage days: 10 -> 20

Risk tolerance: Medium -> High

Storage cost: Rs. 120 -> Rs. 180

(The recommendation should change accordingly.)

Deliverables

D1	Mathematical formulation
D2	Decision variables + constraints
D3	Baseline decision algorithm
D4	Risk-adjusted engine
D5	Sell/Store/Split recommendation
D6	What-if simulator engine

Don't Own (Out of Scope)

- ML forecasting
- Frontend
- Database infrastructure